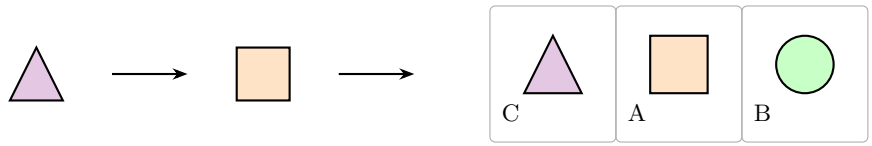
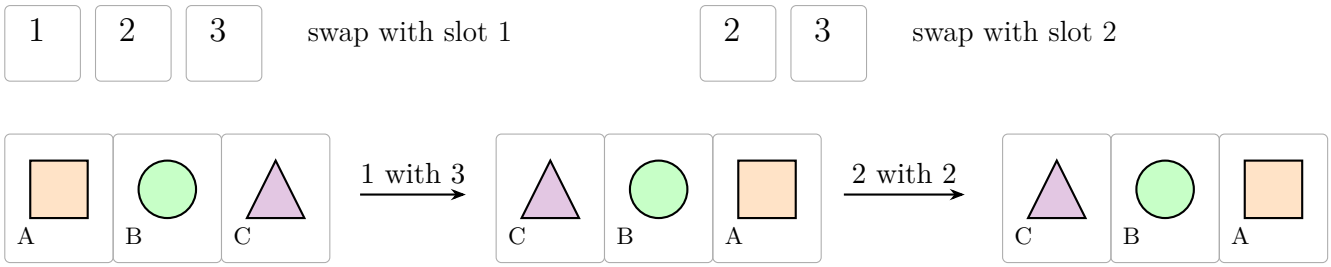
Mix the three picture cards in a cup. Each card has the same chance. Draw the first without looking and keep it out. Draw the second from the two left. Put the last card at the end. Return all cards for a new row.



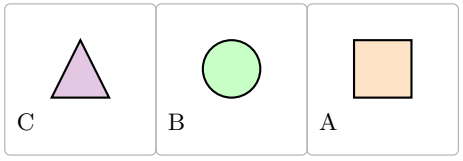
Problem 1: Does this chooser give every order of A, B, C the same chance?

Start A B C. Mix tickets 1, 2, 3 in the cup; draw and swap its slot with slot 1. Empty the cup, then mix tickets 2, 3; draw and swap its slot with slot 2. At each draw, all tickets in the cup have equal chances. Reset both sets for every new row. A slot may swap with itself.



Problem 2: Does this two-swap rule give every order the same chance?

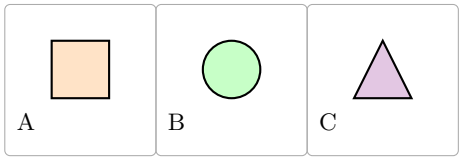
Problem 3: Start with C B A instead. Explain whether the chance of each final order changes, without listing every story again.



A new rule starts with A B C. Draw from 1, 2, 3 at each step: swap slot 1 with its choice, then slot 2 with its choice, then slot 3 with its choice. Replace and mix after each choice.

Problem 4: Can this rule give every order the same chance? Find an exact reason.

Problem 5: Another rule forbids swapping a slot with itself. Swap slot 1 with slot 2 or 3, then swap slots 2 and 3. Find every possible final order. Is this a fair shuffle of all six orders?



Problem 6: Design a four-card ticket-and-swap shuffle on paper that gives every order the same chance, with exactly one choice story per order. Explain why it meets both goals.



Problem 7: Design a fair five-card version on paper with exactly one story for each final order. Specify the ticket sets and resets; how many equally likely stories does your rule have?

